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CS-560: Spring 2026
Final Project Alpha Release V1.2

Alpha Release: Analyzing Data Breach Trends

Basic Information

Project Title: Washington State Data Breach Analysis & Visualization

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GitHub Repository (Code): [CS-560 Final Project \(Chemero\)](#)

GitHub Repository (GitHub Pages): [CS-560 GitHub Pages](#)

Background and Motivation

In an increasingly digital world, data privacy has become a paramount concern for citizens and legislators alike. Washington State requires organizations to report data breaches that affect its residents, creating a rich dataset for public analysis. My motivation for this project stems from a desire to understand the evolving landscape of cybersecurity. By visualizing this data, I hope to uncover whether data breaches are becoming more frequent, which industries are most vulnerable, and how long it typically takes for an organization to notify the public after a breach occurs.

Project Objectives

The primary goal of this visualization is to allow users to explore the severity and nature of data breaches in Washington.

Objective 1: Temporal Trends

Users will be able to identify if the frequency and scale (number of residents affected) of breaches are increasing over time.

Objective 2: Breach Categorization

Users will be able to determine which types of breaches (e.g., Hacking, Phishing, Unauthorized Access) are most prevalent.

Objective 3: Risk Assessment

Users will be able to identify which types of personal information (SSN, Health info, Financial data) are most frequently compromised.

Objective 4: Accountability Analysis

Users will be able to visualize the "Notification Gap" which is the time elapsed between the date of the breach and the date residents were notified.

Data

The project will utilize the dataset: Data Breach Notifications Affecting Washington Residents which is approximately ~6,800 rows and covers the 2017 to 2025 time range.

Format: CSV/JSON

Key Fields:

- Organization Name
- Breach Start/End Dates
- Date of Notification
- Number of WA Residents affected
- Type of Breach
- Information Compromised

Limitations & Considerations:

- The dataset is self reported, which may introduce reporting bias or inconsistencies.
- Some records contain missing or "Unknown" values, particularly in the "Number of Residents Affected" field.
- Date fields may be inconsistent or incomplete, requiring normalization before time series analysis.
- The dataset is limited to Washington State residents, which may affect generalizability to broader cybersecurity trends.

Data Processing

Cleaning: The dataset will require significant cleaning regarding date formats to ensure they are ISO compliant for time series analysis. I will also need to handle "Unknown" or missing values in the "Number of Residents" column, perhaps by imputing the median or labeling them as "Not Reported."

Processing Pipeline:

Preprocessing Stage (Python):

- Data cleaning, normalization of date formats, and handling missing values
- Calculation of derived fields such as Notification Lag and Info Density
- Export of a cleaned and structured dataset for visualization

Visualization Stage (D3.js and JavaScript):

- Dynamic filtering and aggregation
- Real time interaction handling (e.g. brushing, category selection)
- Rendering of visual components and transitions

Derived Attributes:

Notification Lag: A calculated field representing Notification Date to Breach Start Date.

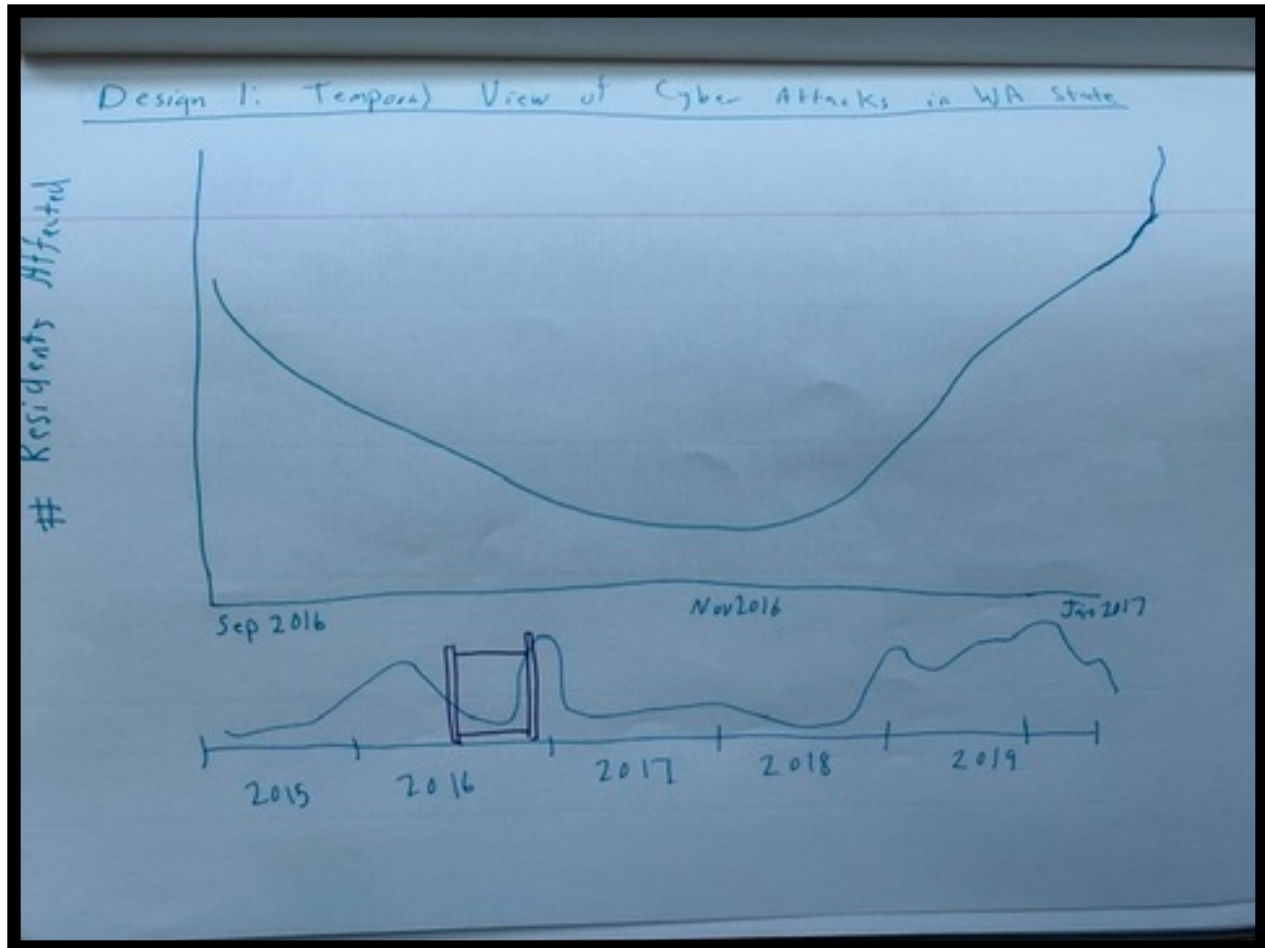
Category Aggregates: Grouping specific breach descriptions into broader categories (e.g., "Cyber Attack," "Human Error," "Physical Theft").

Information Density: A count of how many different types of personal data (SSN, DL, etc.) were stolen per incident.

Visualization Design

Design 1: The Temporal Dashboard

A focus on a large Interactive Area Chart showing the number of residents affected over the last 5 years, with a "brushing" feature to zoom into specific months.



Design 2: The Categorical Network

A node link diagram showing the relationship between "Organization Type" and "Information Stolen."



Design 3: The Heatmap View

A visualization focusing on when breaches occur and the "Notification Lag" intensity, helping to see if certain times of year are more prone to attacks.

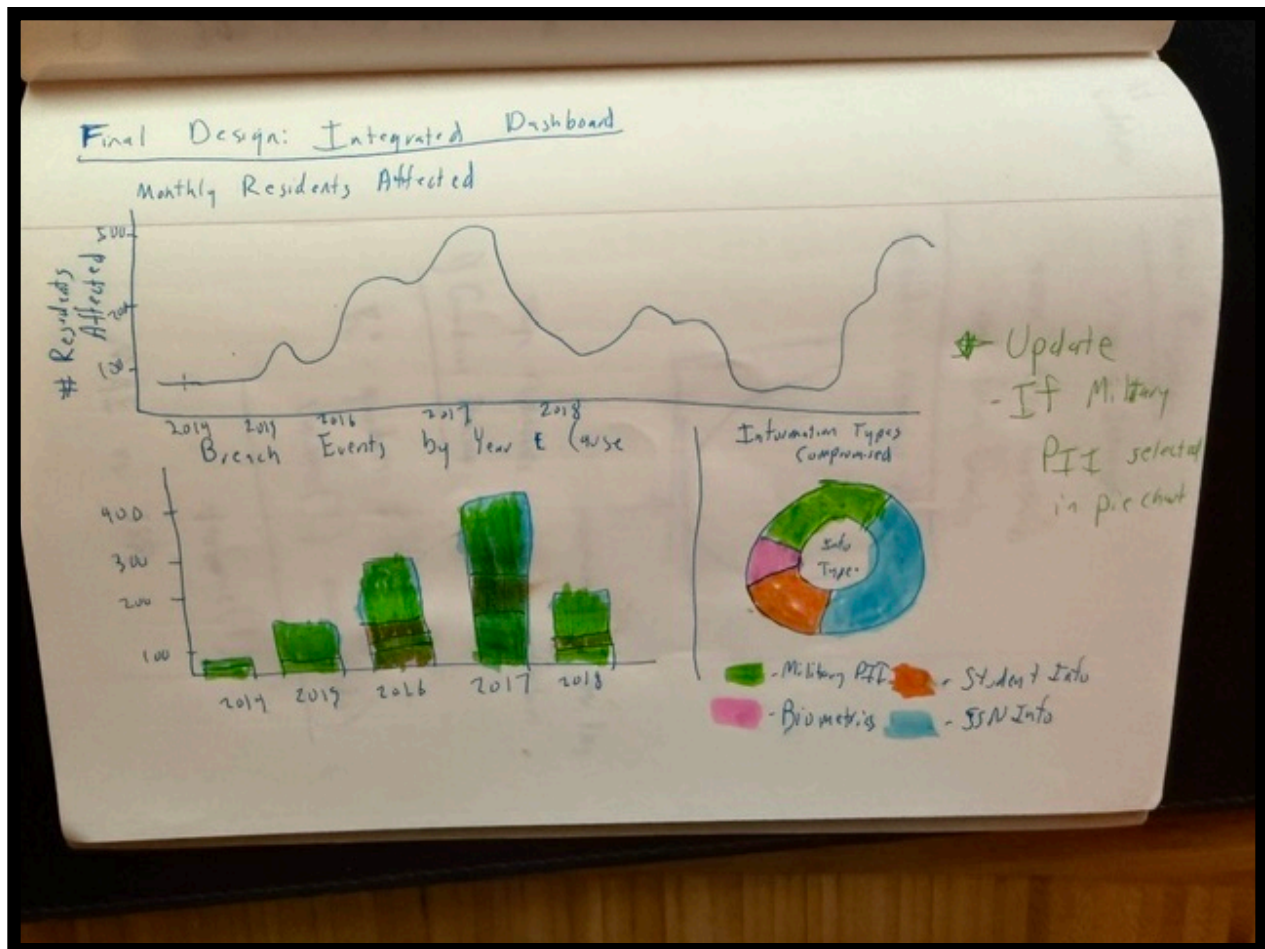


Final Design: Integrated Dashboard (Updated)

The final design will be a Dashboard Interface. It will feature a top level Line Chart for temporal trends (Objective 1). Below that, a Grouped Bar Chart will compare Breach Types (Objective 2). On the right, an interactive Donut Chart will display the breakdown of compromised data types (Objective 3). Selecting a bar or slice will filter the entire dashboard to show data specific to that category.

Update

After discussing this with the professor, we decided to change the stacked bar charts to be individual bar charts that filter depending on which dataset is selected in the pie chart. See the updated image below.



Justification:

This layout allows for high level overview and deep dive filtering, ensuring all four objectives are met within a single view without overwhelming the user.

Must-Have Features

Interactive Timeline:

Users can filter data by year/month. (Meets Obj 1)

Categorical Filter:

A way to toggle between breach types (Hacking vs. Phishing). (Meets Obj 2)

Search Functionality:

A search bar to look up specific organizations (e.g., "Zillow") to see their breach history. (Meets Obj 1 & 4)

Notification Delay Histogram:

A visualization showing the distribution of how long companies take to report breaches. (Meets Obj 4)

Optional Features

Geo Spatial Map:

Layout breaches on a map to show the corporations compromised and where they are located

Extrapolation out into Future (Forecast) of Expected Incidents:

Forecast the future cyber incidents based on visualization trending in temporal and industry analysis.

Project Schedule

Week 1: Research, Design, & Data Preprocessing

Deliverables:

- Five Design Sheet sketches
- Cleaned dataset

Tasks:

- **Apr 4:**
 - Finalize project scope and objectives
 - Review dataset structure and identify key variables
- **Apr 5–6:**
 - Perform Exploratory Data Analysis using Python
 - Identify inconsistencies in "Notification Date" and "Breach Start" fields
- **Apr 7–8:**
 - Clean dataset (handle nulls, normalize categories, standardize date formats)
 - Compute derived fields (Notification Lag, Info Density)
- **Apr 9:**
 - Create and refine Five Design Sheet sketches
 - Initialize GitHub repository and D3.js development environment

Week 2: Frontend Scaffolding & Core Features

Deliverables:

- Revised Proposal, Related Work, and Website
- Functional dashboard layout
- Initial interactive charts

Tasks:

- **Apr 10:**
 - Revised Proposal, Related Work & Website
 - Submit revised proposal and related work documentation
 - Begin HTML/CSS layout (dashboard structure, chart containers)
- **Apr 11:**
 - Continue HTML/CSS layout; publish project website
- **Apr 12–13:**
 - Implement Interactive Timeline (area/line chart)
 - Bind cleaned dataset to visualization
- **Apr 14–15:**
 - Develop categorical visualizations (bar chart / donut chart)
 - Implement filtering logic across components
- **Apr 16:**
 - Validate interactivity and ensure consistent data updates across views

Week 3: Advanced Interactivity & Visual Polishing

Deliverables:

- Search functionality
- Notification delay histogram

- Polished UI/UX
- Alpha Release

Tasks:

- **Apr 17:**
 - Alpha Release
 - Package and submit Alpha Release (functional dashboard with core charts)
 - Begin search/autocomplete functionality for organizations
- **Apr 18:**
 - Complete search/autocomplete implementation
- **Apr 19–20:**
 - Build Notification Delay Histogram
 - Validate correctness of lag calculations
- **Apr 21–22:**
 - Refine visual encodings (color schemes, labels, tooltips)
 - Improve responsiveness and layout
- **Apr 23:**
 - Conduct usability testing and incorporate feedback

Week 4: Deployment, QA, & Final Deliverables

Deliverables:

- Beta Release

Tasks:

- **Apr 24–25:**
 - Perform QA testing (cross-browser compatibility, edge cases)
- **Apr 26–27:**
 - Finalize documentation (GitHub README, architecture overview)
- **Apr 28–29:**
 - Begin final report (design decisions, outcomes, limitations)
- **Apr 30:**
 - Beta Release
 - Submit Beta Release

Week 5: Presentation Preparation

Deliverables:

- Project Presentation

Tasks:

- **May 1–4:**
 - Build presentation slides covering design rationale, key findings, and live demo
- **May 5–9:**
 - Rehearse presentation; incorporate advisor feedback
- **May 10:**
 - Final dry run; polish slides and demo flow
- **May 11:**
 - Submit/deliver Project Presentation

Week 6: Report Draft

Deliverables:

- Project Report Draft

Tasks:

- **May 12–14:**
 - Write report sections: Introduction, Related Work, Design Decisions
- **May 15–16:**
 - Write Outcomes, Limitations, and Future Work sections
- **May 17:**
 - Submit Project Report Draft

Week 7: Final Submission

Deliverables:

- Final Submission

Tasks:

- **May 18:**
 - Incorporate feedback from report draft; record video demo
- **May 19:**
 - Final polish; dashboard, report, and video
- **May 20:**
 - Submit Final Submission package

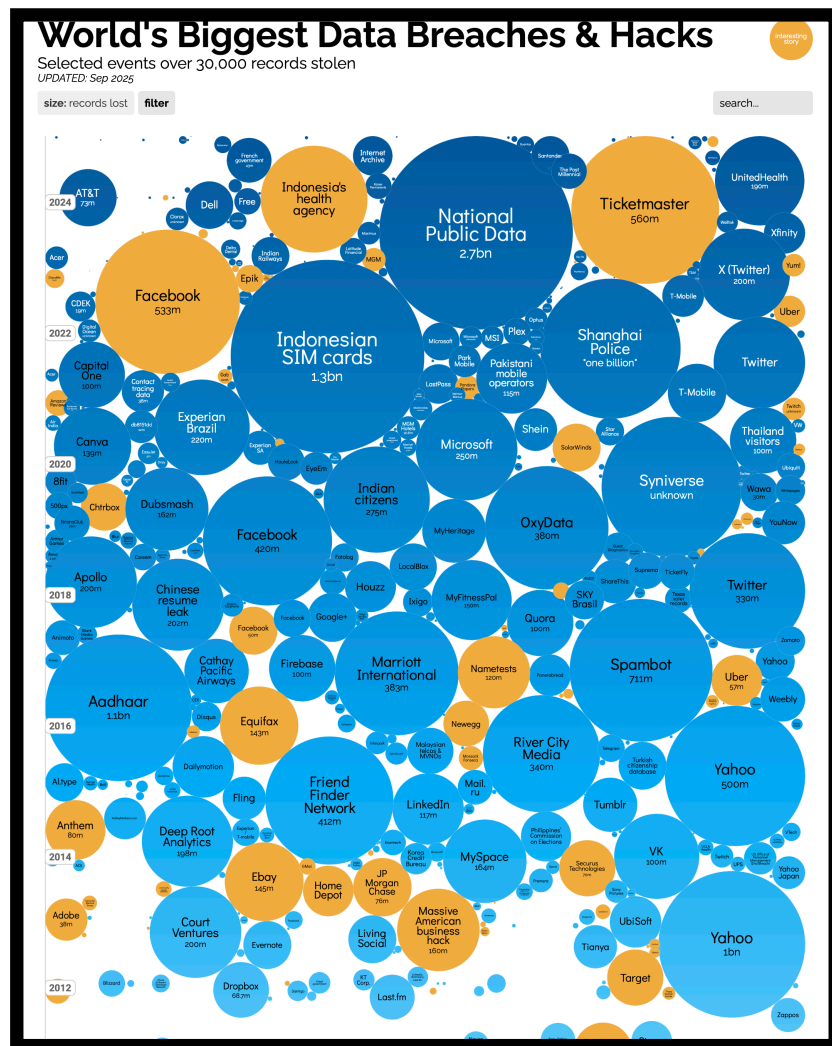
Related Works

Resource #1: Ranking of World's Largest Data Breaches

Source: <https://informationisbeautiful.net/visualizations/worlds-biggest-data-breaches-hacks/>

This site provides a really nice example of how to visualize data breaches. I'm not doing this chart, however, it is definitely an eye catcher. The graphic ranks the largest data breaches using a bubble chart, with additional encoding via bubble color. You can interact with the bubbles by clicking on the yellow ones.

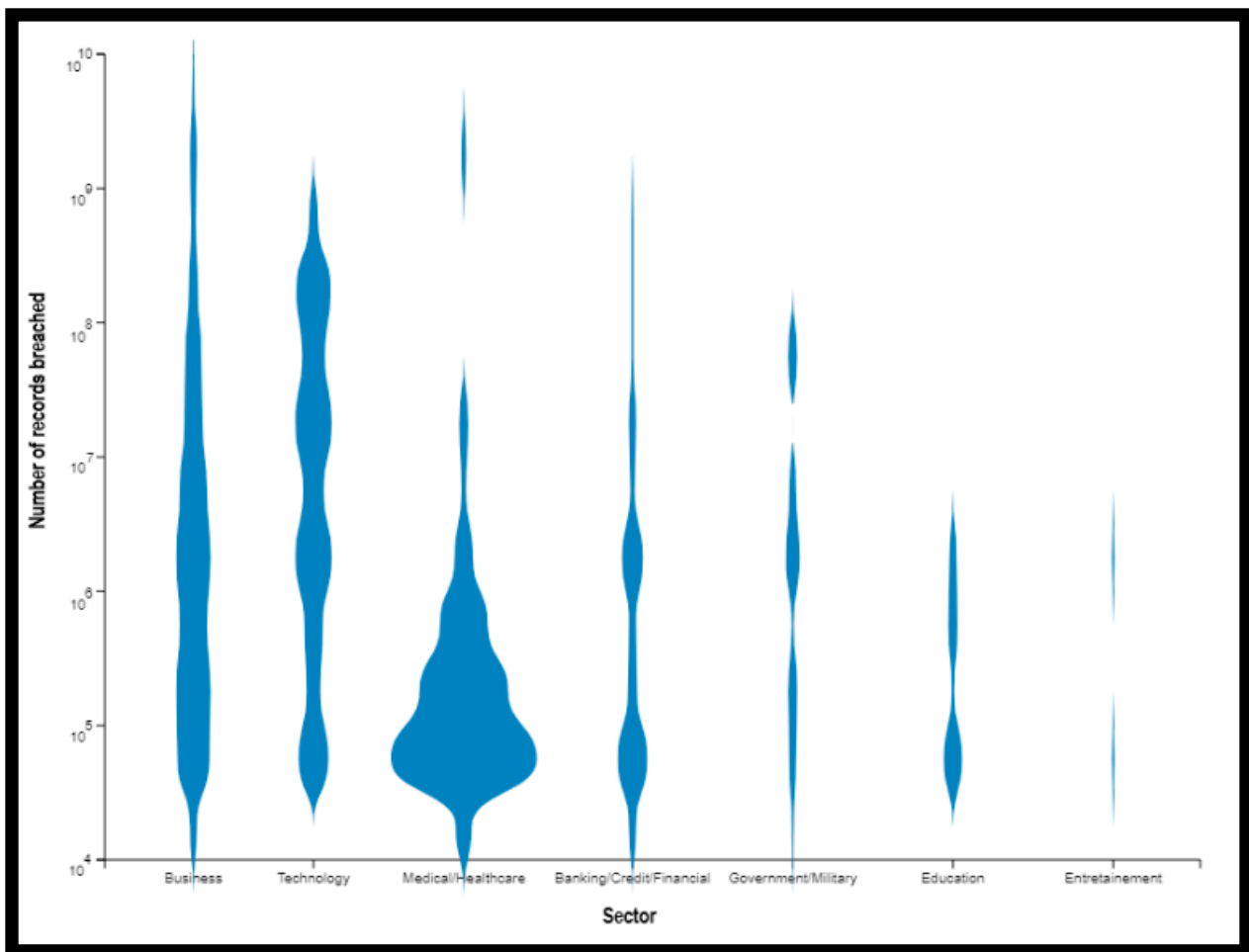
One example is the SolarWinds breach, which I was lucky enough to experience first hand. In 2020, Russian threat actors compromised the security company SolarWinds and used that foothold to extend their control into government and private enterprises both domestically and abroad. It was...fun, to say the least.



Resource #2: Understanding Data Breach from a Global Perspective

Source: <https://www.mdpi.com/2306-5729/9/2/27>

This is another great source that provides numerous examples of how one can visualize data breach incidents. In this article there are pie charts, geographic graphs, proportional sum (pie) charts, box plot charts, Sankey diagrams and even a few charts I don't think we covered in class. The takeaway from this paper is that there are many ways to visualize the dataset that I'm interested in. A good example is the violin plot below showing number of records stolen per sector:



Resource #3: Patterns in State Level Data Breach Reporting

Source: <https://arxiv.org/pdf/2310.04786>

This article is unique because it studied the regulatory reporting requirements and breach reports for US states. Washington, which is the dataset that I'm using for my project, is specifically called out in the report because it has much more rigid reporting requirements. This paper is relevant because it helps readers of my analysis understand that they may not be able to get the same statistics for other states. For example, only 14 states actually have mandatory reporting and not every element of a breach is reportable. There is drastic deviation in the

Breach Notice, Breach Start Date and Breach End Date reporting requirements. See below for a high level breakdown:

Table 1: Definitions of reportable data breaches in the U.S. in 2021	
Notification to state Attorney General	Number of states ^a
No obligations	17
Yes	14
Yes if more than 250 state residents	4
Yes if more than 500 state residents	8
Yes if more than 1000 state residents	7
Others	4
^a including 50 states, District of Columbia, Guam, Puerto Rico, and the Virgin Islands	

Credit to Benjamin Avanzi, Xingyun Tan, Greg Taylor, Bernard Wong for these informatics.

Resource #4: An Exploratory Data Analysis of Data Breaches Over Time

Source: <https://www.sciencedirect.com/science/article/pii/S1877050919306064>

This paper is a deep dive on another breach dataset. This dataset is much larger, with 11.5 billion stolen records being tracked across nine thousand incidents. The data goes back to 2005 so it is somewhat of a superset of my dataset. My Washington state data ranges from 2017 to 2025 so it is a much smaller pool than what is covered here.

I like this because I can compare my findings against this paper and hopefully generate some macro level themes. For instance, are Washington state breaches worse than the national average? Are they more frequent? Is the threat actor the same? These are all direct comparisons that this paper allows me to do.

Resource #5: Analyzing the Long Tail on Breach Data

Source: <https://academic.oup.com/cybersecurity/article/2/1/3/2736315?login=false>

This is another fascinating angle on visualizing data breaches. This article delves into the unique data anomalies that we typically encounter when analyzing breach data. The point of the paper is that outliers in these datasets are common and they're heavy. The paper gets pretty specific on the data shape expectations (normalization) but the main takeaway is to interrogate outliers and double check that they're not causing visualizations to render in a misleading way.

Delivered Work

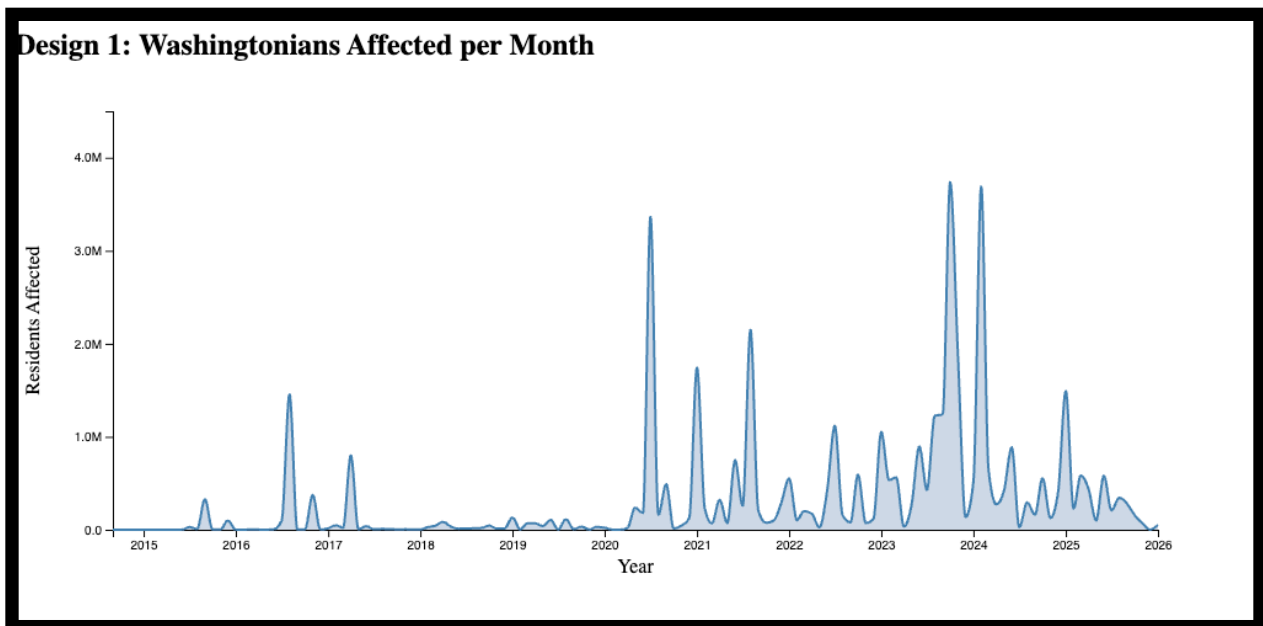
See the list of completed items thus far. You can see I'm currently on track with the deliverables stated in the roadmap.

Week 1

The following deliverables were completed during week 1 of the project.

- Finalized project scope and objectives
- Performed exploratory data analysis in Python
- **Cleaned the dataset:**
 - Normalized date formats, handled null/unknown values in the Residents Affected field, and standardized categorical fields
- **Computed derived fields:**
 - Notification Lag and Information Density
- Started GitHub repository and D3.js development environment

The following design was also created towards the end of my week 1 sprint. This was implemented using D3 and is visible on the project website. This is currently in a rough draft format and there isn't any interaction or CSS around the graph.

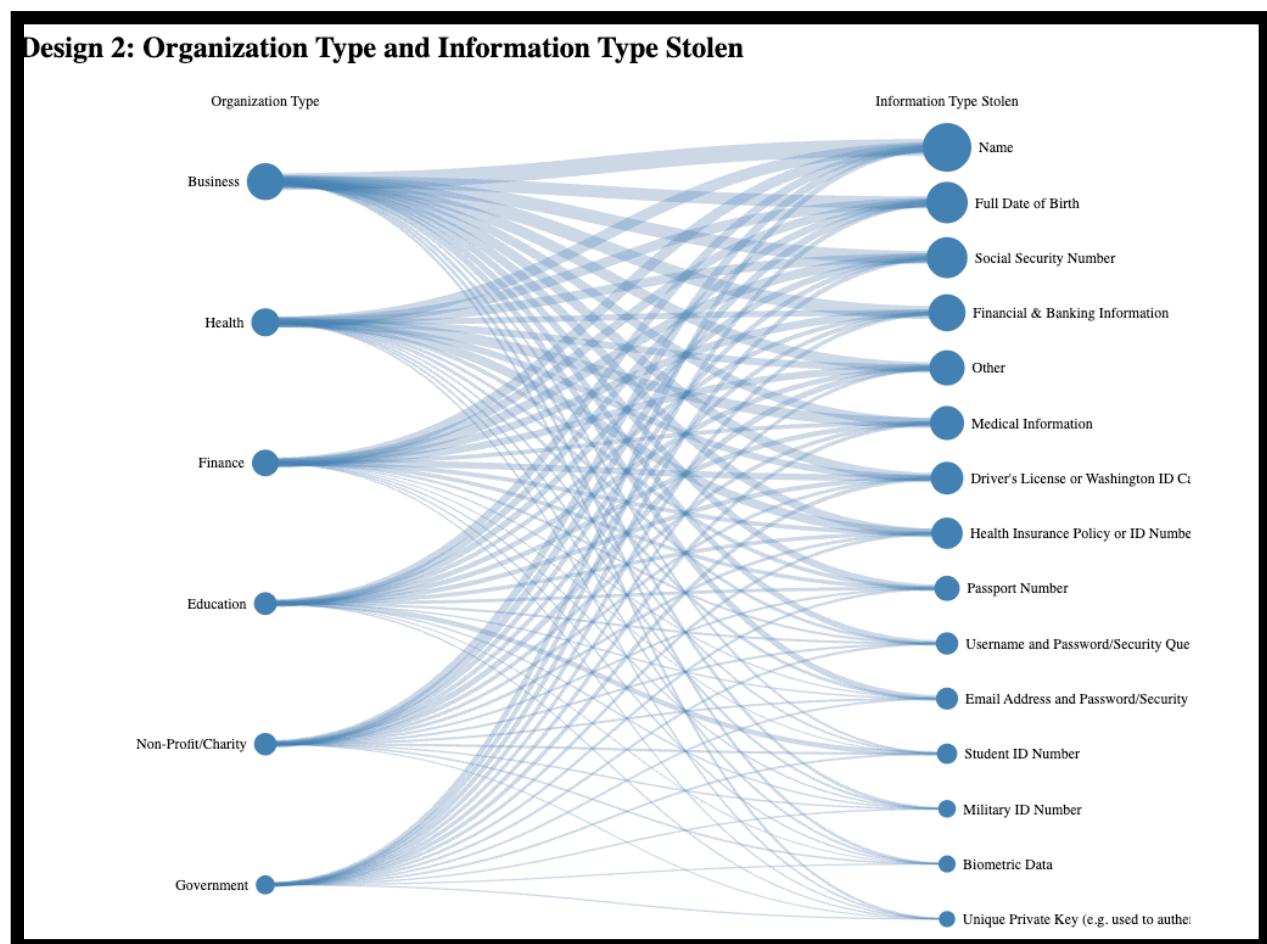


Week 2

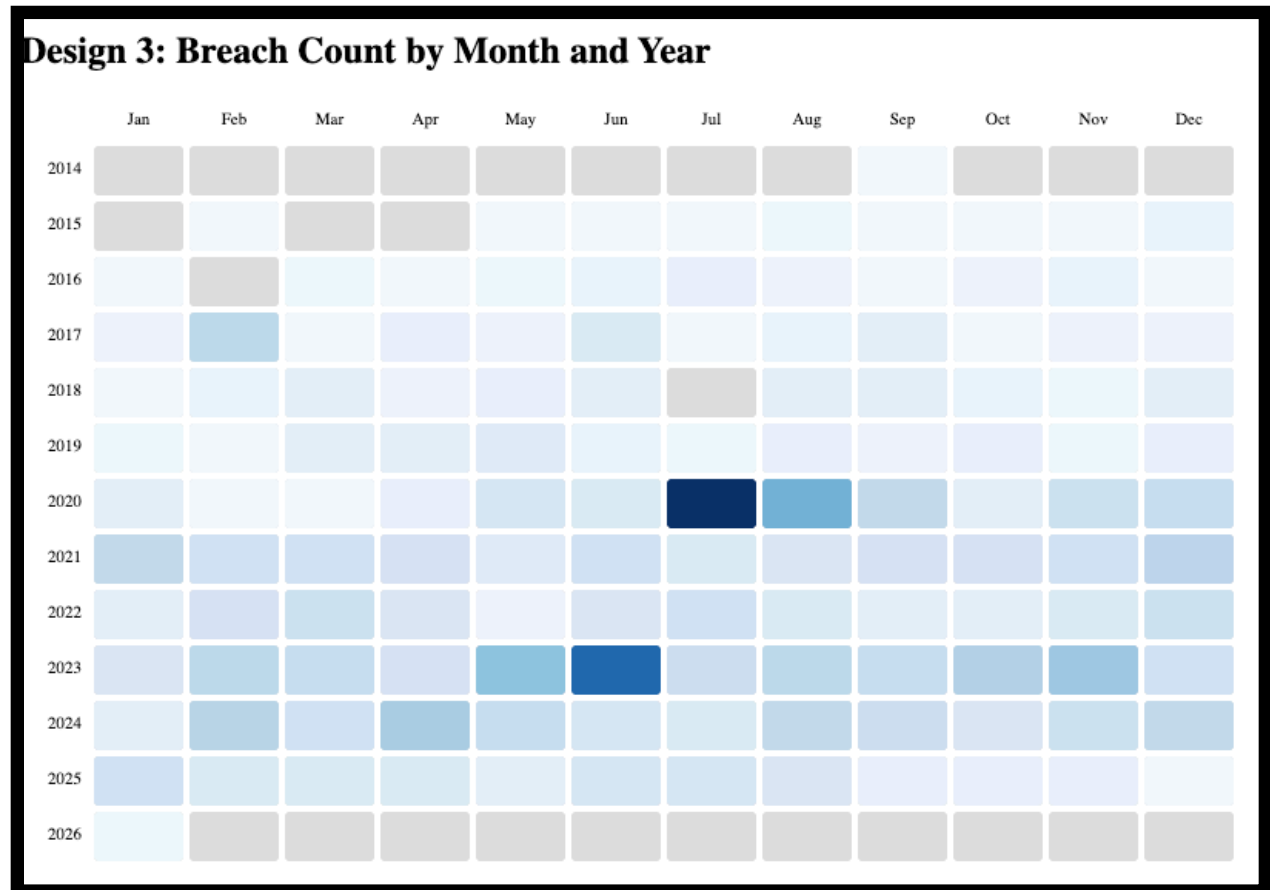
The following deliverables were completed during week 2 of the project.

- Created a basic version of charts 2, 3 and the final
 - Currently the charts don't have any tooltips or fancy formatting
- Submitted revised proposal, related work documentation, and project website
- Worked on the interactive timeline (line/area chart) bound to the cleaned dataset
 - This part is still not working but it isn't required by the end of Alpha release
- Developed initial categorical visualizations (bar chart and donut chart)

Design 2

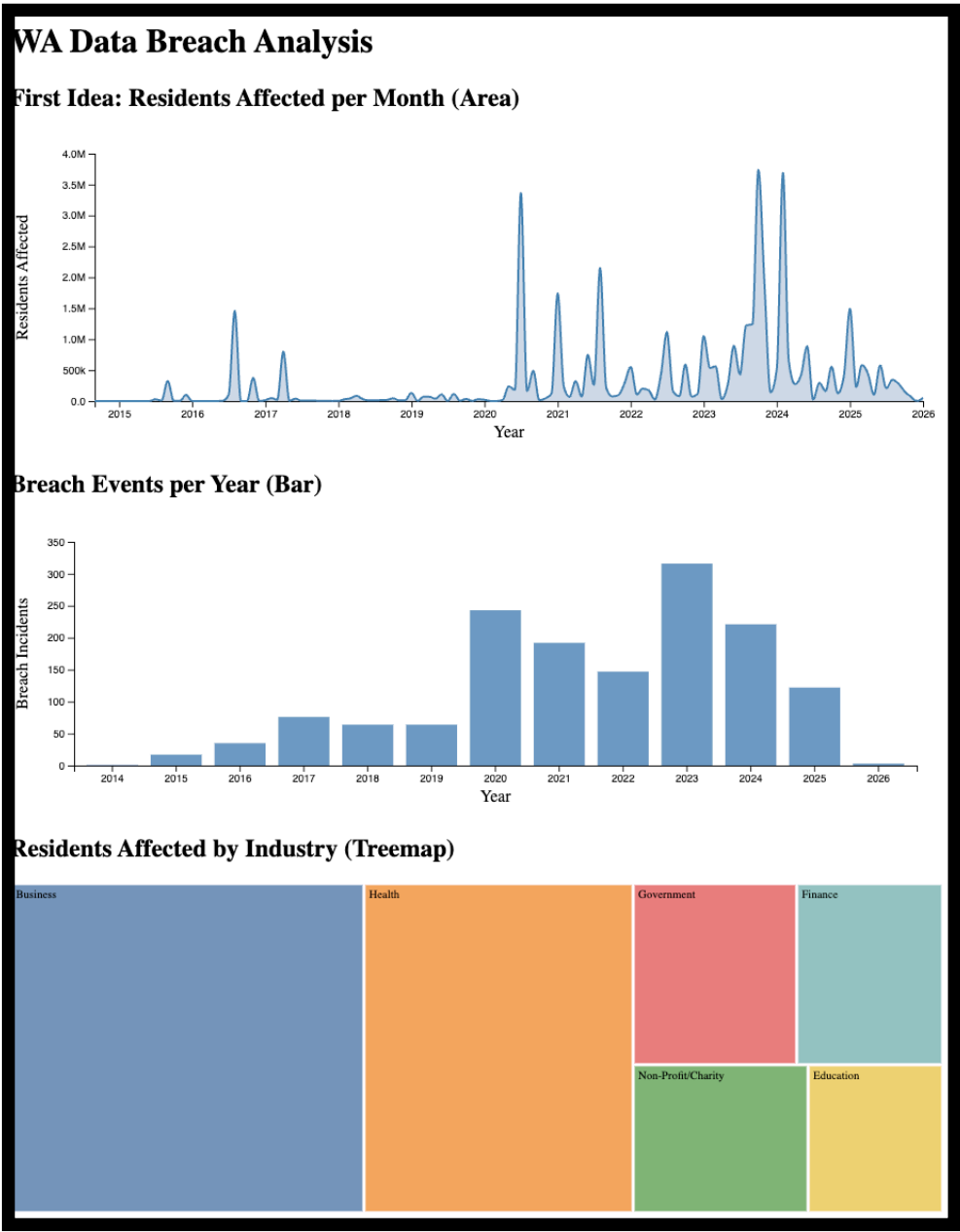


Design 3



Final Design (Slightly Different for Proposal)

My current version is slightly different from what I initially drafted. When creating the pie chart I found that it didn't look that nice because of the spacing. I instead tried a heat map to see if that would fit better with the layout. I'm not sure yet which way I'm leaning on this one.



Related Works

At this point in the projects there are no roadblocks. The chart basics have been implemented without too much difficulty. I think for the final version I'll need to do a lot of CSS work to make the it look sharp. Overall, no concerns with the rest of the deliverables.

Works Cited

Data Sources

McCandless, D. (n.d.). World's biggest data breaches & hacks [Interactive visualization]. Information Is Beautiful. <https://informationisbeautiful.net/visualizations/worlds-biggest-data-breaches-hacks/>

U.S. General Services Administration. (n.d.). Data breach notifications affecting Washington residents. Data.gov. <https://catalog.data.gov/dataset/data-breach-notifications-affecting-washington-residents-personal-information-breakdown>

Visualization & Design Methodology

Five Design Sheet Methodology. (n.d.). The Five Design Sheet methodology. <https://fds-design.github.io/>

Roberts, J. C., Headleand, C., & Ritsos, P. D. (2016). Sketching designs using the Five Design Sheet methodology. IEEE Transactions on Visualization and Computer Graphics.

Related Research

Domain Context (Cybersecurity & Data Breaches)

Avanzi, B., Tan, X., Taylor, G., & Wong, B. (2023). On the evolution of data breach reporting patterns and frequency in the United States: A cross-state analysis. arXiv. <https://arxiv.org/pdf/2310.04786>

Edwards, B., Hofmeyr, S., & Forrest, S. (2016). Hype and heavy tails: A closer look at data breaches. Journal of Cybersecurity, 2(1), 3–14. <https://doi.org/10.1093/cybsec/tyw003>

Hammouchi, H., Cherqi, O., Mezzour, G., Ghogho, M., & El Koutbi, M. (2019). Digging deeper into data breaches: An exploratory data analysis of hacking breaches over time. Procedia Computer Science, 151, 1004–1009. <https://doi.org/10.1016/j.procs.2019.04.141>

Identity Theft Resource Center. (2023). Annual data breach report. <https://www.idtheftcenter.org/>

Rodrigues, G. A. P., Serrano, A. L. M., Lemos, A. N. L. E., Canedo, E. D., de Mendonça, F. L. L., de Oliveira Albuquerque, R., Orozco, A. L. S., & García Villalba, L. J. (2024). Understanding data breach from a global perspective: Incident visualization and data protection law review. Data, 9(2), 27. <https://doi.org/10.3390/data9020027>

Verizon. (2023). Data breach investigations report (DBIR). <https://www.verizon.com/business/resources/reports/dbir/>

AI Usage Statement

This project utilized AI assisted tools, including Gemini, to support formatting, organization, and refinement of written content. All substantive ideas, design decisions, data analysis plans, and implementation details were developed independently by the author. AI tools were used solely to improve clarity and presentation.